TEACHER PACKET · PRE-CALCULUS · WED · –

Multiplying and Dividing Rational Expressions

Multiply and divide rational expressions, simplify, and state restrictions.

01 · PROJECT

Today's addition — Multiply & Divide Rational Expressions

NEW CONCEPT

1 / 4

Same four moves, every problem

- **Factor** every top and bottom completely (GCF, difference of squares, trinomials).
- **Dividing?** Flip the second fraction FIRST — then it's a multiply problem.
- **Cancel** matching factors, then simplify.
- **Restrictions** come from the ORIGINAL denominators — before anything cancels.

WORKED EXAMPLE

2 / 4

We do: divide with a trinomial

WORKED EXAMPLE

$\dfrac{2x^2+9x+4}{3x} \div \dfrac{2x+1}{x}$

1. Flip: $\cdot \dfrac{x}{2x+1}$.

2. Factor: $2x^2+9x+4=(2x+1)(x+4)$.

3. Cancel $(2x+1)$ and x: $\dfrac{x+4}{3}$.

4. Restrictions: $x \neq 0, -\frac{1}{2}$.

WATCH OUT

The restriction trap

COMMON MISTAKE

Restrictions come from the ORIGINAL denominators — including factors that later cancel, and the divisor's TOP when you divide.
If you read them off the simplified answer, you miss values like $x = -2$ in Example 1.

KEEP IN MIND

Write the restrictions down at the FACTOR step, before you cancel anything.

YOU TRY

Your turn

YOU TRY

Simplify and state restrictions:
(a) $\frac{8m}{m^2 + 5m} \cdot \frac{m}{4}$
(b) $\frac{4p}{p^2} \div \frac{2}{p}$

02 · MODEL ON THE BOARD

Worked procedure — “I do”

1 **Simplify** $\dfrac{x^2+5x+6}{x} \cdot \dfrac{x}{x+2}$ **and state restrictions.**

1. Factor: $x^2 + 5x + 6 = (x+2)(x+3)$.
2. Write as one fraction: $\dfrac{(x+2)(x+3)\cdot x}{x\cdot(x+2)}$.
3. Cancel $(x+2)$ and x.
4. State restrictions from the original denominators: $x \neq 0, -2$.

ANSWER

$x+3,\ x \neq 0, -2$

2 **Simplify** $\dfrac{2x^2+9x+4}{3x} \div \dfrac{2x+1}{x}$ **and state restrictions.**

1. Multiply by the reciprocal: $\dfrac{2x^2+9x+4}{3x} \cdot \dfrac{x}{2x+1}$.
2. Factor: $2x^2 + 9x + 4 = (2x+1)(x+4)$.
3. Cancel $(2x+1)$ and x: $\dfrac{x+4}{3}$.
4. Restrictions: $3x \neq 0$, divisor top $2x+1 \neq 0$, divisor bottom $x \neq 0 \rightarrow x \neq 0, -\frac{1}{2}$.

ANSWER

$\dfrac{x+4}{3},\ x \neq 0, -\frac{1}{2}$

03 · HANDOUT

Guided notes — teacher key (students get blanks, printed 2-up)

Multiplying & Dividing Rational Expressions

TEACHER KEY · —

Formula bank **Factor toolbox:** GCF · difference of squares $a^2 - b^2 = (a-b)(a+b)$ · trinomials

Divide rule: $\frac{A}{B} \div \frac{C}{D} = \frac{A}{B} \cdot \frac{D}{C}$ (flip, then multiply)

Restrictions: every ORIGINAL denominator $\neq 0$ — and when dividing, the divisor's top too

VOCABULARY

Rational expression — a fraction whose top and bottom are **polynomials**

Restriction — an x-value that makes an ORIGINAL **denominator** equal zero

Reciprocal — the flipped fraction — dividing means multiplying by the **reciprocal**

EXAMPLE 1 · TYPE 1 — MULTIPLY

Simplify $\frac{x^2 + 5x + 6}{x} \cdot \frac{x}{x+2}$ **and state restrictions.**

1. Factor every top and bottom: $x^2 + 5x + 6 = (x+2)(x+3)$
2. Cancel common factors: the $(x+2)$ pair and the x pair cancel
3. Simplified result: $x + 3$
4. Restrictions from ORIGINAL denominators: $x \neq 0, -2$

$x + 3$, $x \neq 0, -2$

EXAMPLE 2 · TYPE 2 — DIVIDE (FLIP FIRST)

Simplify $\frac{2x^2 + 9x + 4}{3x} \div \frac{2x+1}{x}$ **and state restrictions.**

1. Flip the divisor and multiply: $\div \frac{2x+1}{x}$ becomes $\cdot \frac{x}{2x+1}$
2. Factor: $2x^2 + 9x + 4 = (2x+1)(x+4)$
3. Cancel $(2x+1)$ and x: result $\frac{x+4}{3}$
4. Restrictions (original denominators AND the divisor's top): $x \neq 0, -\frac{1}{2}$

$\frac{x+4}{3}$, $x \neq 0, -\frac{1}{2}$

YOU TRY

1.

Simplify $\frac{8m}{m^2 + 5m} \cdot \frac{m}{4}$ and state restrictions.

$\frac{2m}{m+5}$, $m \neq 0, -5$

2.

Simplify $\frac{3a}{a^2 + a} \div \frac{3}{a}$ and state restrictions.

$\frac{a}{a+1}$, $a \neq 0, -1$

SUMMARY

Factor → **cancel** → simplify → restrictions from every **original denominator** (flip FIRST when dividing).

04 · PRACTICE

Worksheet — Multiplying and Dividing Rational Expressions · YOURS (mirror)

Factor first, cancel common factors, simplify, and state the restrictions from every ORIGINAL denominator (and the divisor when dividing).

TYPE 1 **Multiply, simplify, and state restrictions**

EXAMPLE

$\dfrac{x^2+5x+6}{x}\cdot\dfrac{x}{x+2}$. Factor: $x^2+5x+6=(x+2)(x+3)$, so $\dfrac{(x+2)(x+3)}{x}\cdot\dfrac{x}{x+2}=x+3$, with $x \neq 0, -2$.

Directions: Multiply, simplify, and state restrictions.

1. $\dfrac{4x}{x^2-16}\cdot\dfrac{x-4}{3}$

2. $\dfrac{8m}{m^2+5m}\cdot\dfrac{m}{4}$

3. $\dfrac{x^2+7x+10}{x}\cdot\dfrac{3x}{x+5}$

4. $\dfrac{2x^2+7x+3}{x}\cdot\dfrac{x}{2x+1}$

5. $\dfrac{6y}{9}\cdot\dfrac{3y^2}{y}$

6. $\dfrac{2x^2}{10x}\cdot\dfrac{5}{x+1}$

TYPE 2 **Divide, simplify, and state restrictions**

EXAMPLE

$\dfrac{2x^2+9x+4}{3x}\div\dfrac{2x+1}{x}$. Factor $2x^2+9x+4=(2x+1)(x+4)$, flip and multiply: $\dfrac{(2x+1)(x+4)}{3x}\cdot\dfrac{x}{2x+1}=\dfrac{x+4}{3}$, with $x \neq 0, -\frac{1}{2}$.

Directions: Divide, simplify, and state restrictions.

7. $\dfrac{x^2-25}{5x}\div\dfrac{x-5}{x}$

8. $\dfrac{x^2+6x+8}{x}\div\dfrac{x+2}{2x}$

9. $\dfrac{3x^2+11x+6}{x}\div\dfrac{3x+2}{x}$

10. $\dfrac{3a}{a^2+a}\div\dfrac{3}{a}$

11. $\dfrac{2y}{y^2-4}\div\dfrac{y+2}{4}$

12. $\dfrac{4p}{p^2}\div\dfrac{2}{p}$

05 · KEY **Answer key**

MULTIPLY, SIMPLIFY, AND STATE RESTRICTIONS

1. $\frac{4x}{3(x+4)}$; restrictions $x \neq 4, -4$

2. $\frac{2m}{m+5}$; restrictions $m \neq 0, -5$ — ⚠ the sheet's key shows $\frac{2}{m+5}$, which drops an m: $\frac{8m}{m(m+5)} \cdot \frac{m}{4} = \frac{8m^2}{4m(m+5)} = \frac{2m}{m+5}$

3. $3(x+2)$; restrictions $x \neq 0, -5$

4. $x+3$; restrictions $x \neq 0, -\frac{1}{2}$

5. $2y^2$; restriction $y \neq 0$

6. $\frac{x}{x+1}$; restrictions $x \neq 0, -1$

DIVIDE, SIMPLIFY, AND STATE RESTRICTIONS

7. $\frac{x+5}{5}$; restrictions $x \neq 0, 5$

8. $2(x+4)$; restrictions $x \neq 0, -2$

9. $x+3$; restrictions $x \neq 0, -\frac{2}{3}$

10. $\frac{a}{a+1}$; restrictions $a \neq 0, -1$ — ⚠ the sheet's key shows $\frac{1}{a+1}$, which drops an a: $\frac{3a}{a(a+1)} \cdot \frac{a}{3} = \frac{3a \cdot a}{3a(a+1)} = \frac{a}{a+1}$

11. $\frac{8y}{(y-2)(y+2)^2}$; restrictions $y \neq 2, -2$ — ⚠ the sheet's key shows $\frac{2}{y-2}$; flipping gives $\frac{2y}{(y-2)(y+2)} \cdot \frac{4}{y+2} = \frac{8y}{(y-2)(y+2)^2}$ and nothing else cancels

12. 2; restriction $p \neq 0$

06 · CLOSE **Exit ticket**

1. Simplify $\frac{x^2-9}{2x} \cdot \frac{x}{x+3}$ and state the restrictions.